

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.
(2 points each)

Asexual fitness advantage
Cost of sex
Ecological release
Gametic conflict
Good genes
Hermaphrodite

Intrasexual selection
Intersexual selection
Niche exploitation
Runaway sexual selection
Sensory bias
Sensory drive

Sexual conflict
Sexual fitness disadvantage
Sneaking females
Sneaking males
Sperm competition
Spermicide

1. _____ In the words of the eminent sponge, Dr. Judd: dash and squirt.
2. _____ One taxon goes extinct, allows another taxon to diversify into the previously occupied niches.
3. _____ Different self-interests of males and females.
4. _____ Toxic seminal fluid reduces swimming ability of sperm from another male.
5. _____ Favorite band or musical artist (not in list above).

- (5 pts) 6. List three of the five major mass extinctions that have occurred since the Cambrian. Include and indicate which event was the largest and which event is the most famous.

- (10 pts) 7. Illustrate and explain the relative fitness of males and of females at different ages or size classes that could lead to the evolution of protandry. Then, explain when relative fitness of males and females favors the evolution of or maintenance of separate sexes (i.e., no hermaphroditism).

8. Male swordtail characin fishes (*Corynopoma riisei*) have a long, narrow extension of the gill cover. The extension has a “flag” on the tip, which mimics the fish’s prey. The male uses the flag to lure females to the male for mating. A study by Arnqvist and Kolm (2010) showed that the shape of the flag was strongly correlated with environmental variables like prey availability. Based on this information, answer the next two questions.

(5 pts) (a) Explain whether this is intrasexual or intersexual selection and why.

(10 pts) (b) Decide whether the flag more likely evolved by sensory bias or sensory drive. Explain what is sensory bias or sensory drive (depending on your choice), and justify your decision by explaining how your choice is consistent with the scenario described above.

- (15 pts) 9. Fossil evidence suggests dinosaurs (excluding birds) had high turnover rates. Describe how ecological specialization and population dynamics might explain high turnover rate in dinosaurs. Include both origination and extinction.

- (15 pts) 10. Male satin bowerbirds decorate their bowers with bright blue objects they find around the area. The bower is not a nest. It's a structure built solely to attract females. Females build the nest elsewhere. Assuming the male does *not* help raise his offspring, explain how the male's ability to gather decorative objects may serve as an indirect indicator of male genetic quality that affects female choice. In your explanation, be sure to state which intersexual selection hypothesis best fits this scenario, and why.

- (20 pts) 11. Discuss in terms of fitness and adaptive flexibility the advantages and disadvantages of both asexual and sexual reproduction. That is, what are the advantages and disadvantages of maintaining the same genome (asexual) versus a variable genome (sexual) in relation to the current environment (in a broad sense) and possible future changes to the environment. Be sure to include and explain the “cost of sex” in your discussion, as it relates to advantages of disadvantages of each mode of reproduction.